

Applications of base editing and prime editing

Matt Coelho

Cancer Research UK Career Development Fellow
Wellcome Sanger Institute

The variant to function problem in cancer

- Cancer genome sequencing is used to direct precision medicine
- Understanding somatic variant function is a major clinical bottleneck, especially in the context of therapy
- Can we predict drug resistance mutations?
 - > Improved patient stratification
 - > Effective second-line therapies
 - > Novel inhibitors of mutated cancer targets

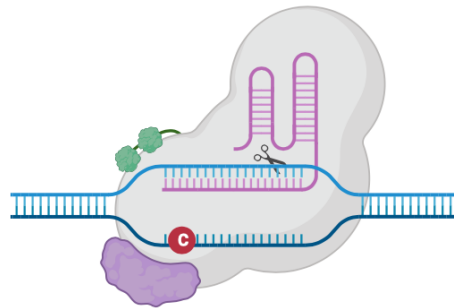
The variant to function problem in cancer

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Functional genomics approach

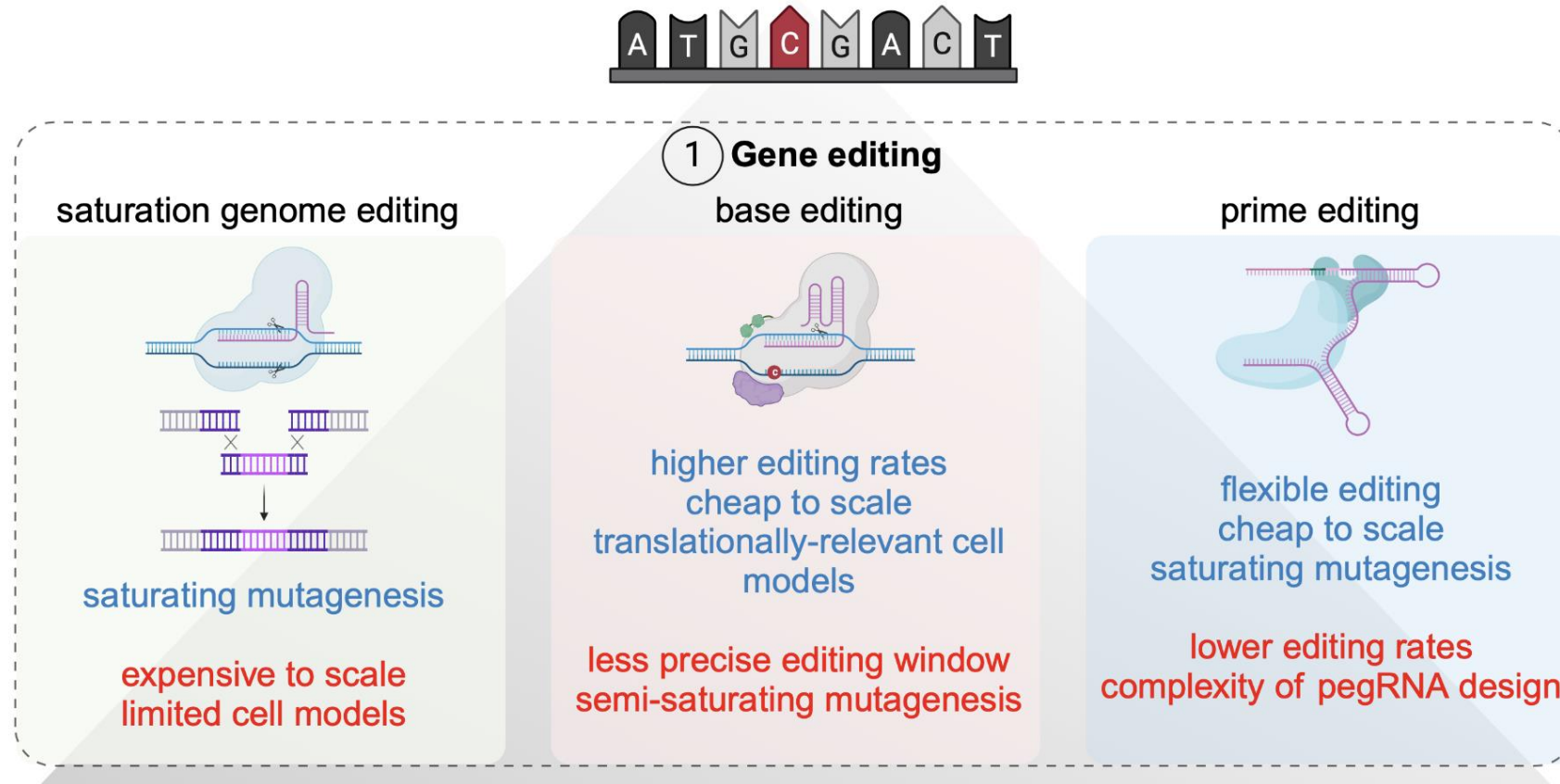
Build a map of variant effect in cancer to help translate genomics data into better treatments for patients



gene editing:

- *scalable*
- *rapid*
- *functional*
- *sensitive*
- *mechanistic insight*

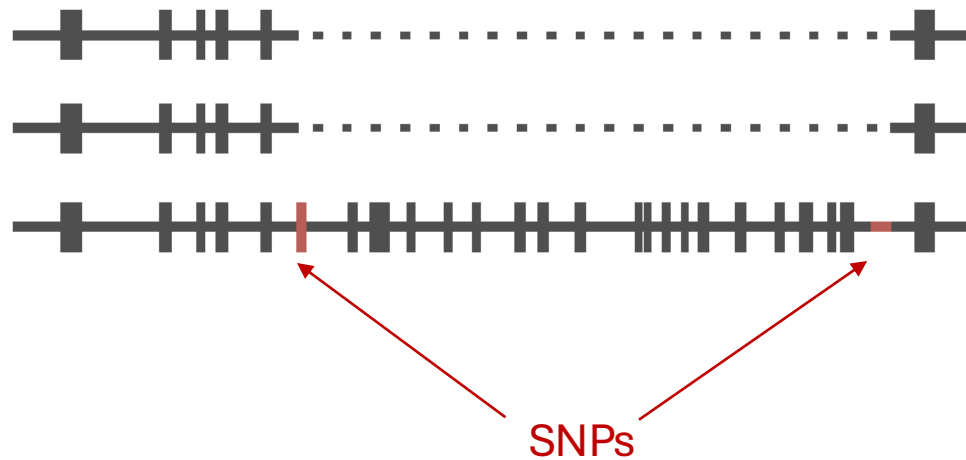
Functional interrogation of DNA variants at scale



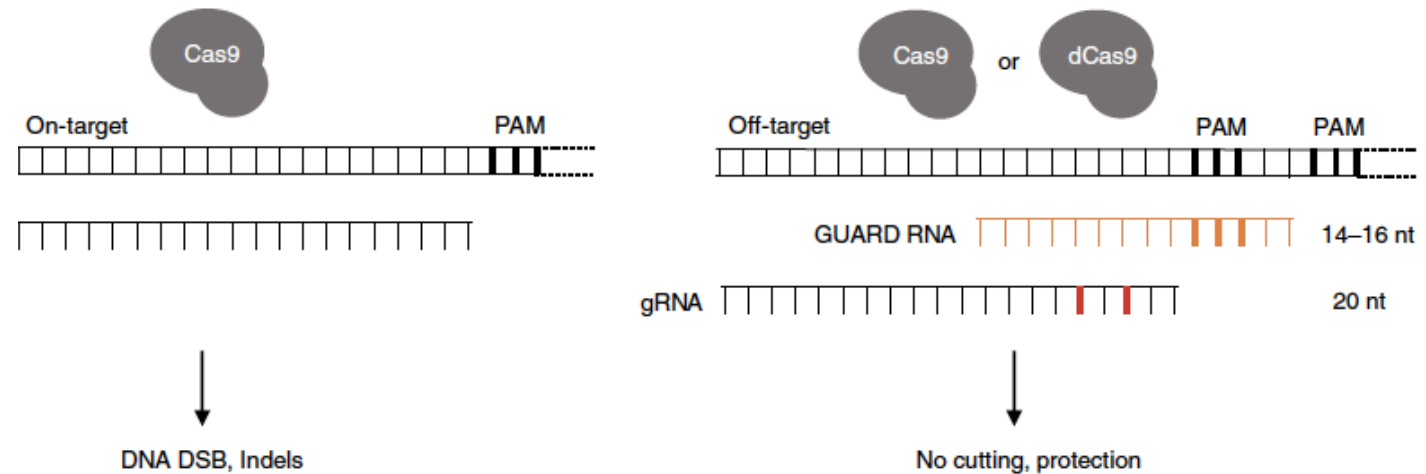
Cooper, Obolenski, Waters, Bassett* & **Coelho***, *Cell Reports Methods*, 2024

Saturation prime editing

NPC1 locus



CRISPR GUARD protects off-target sites from Cas9

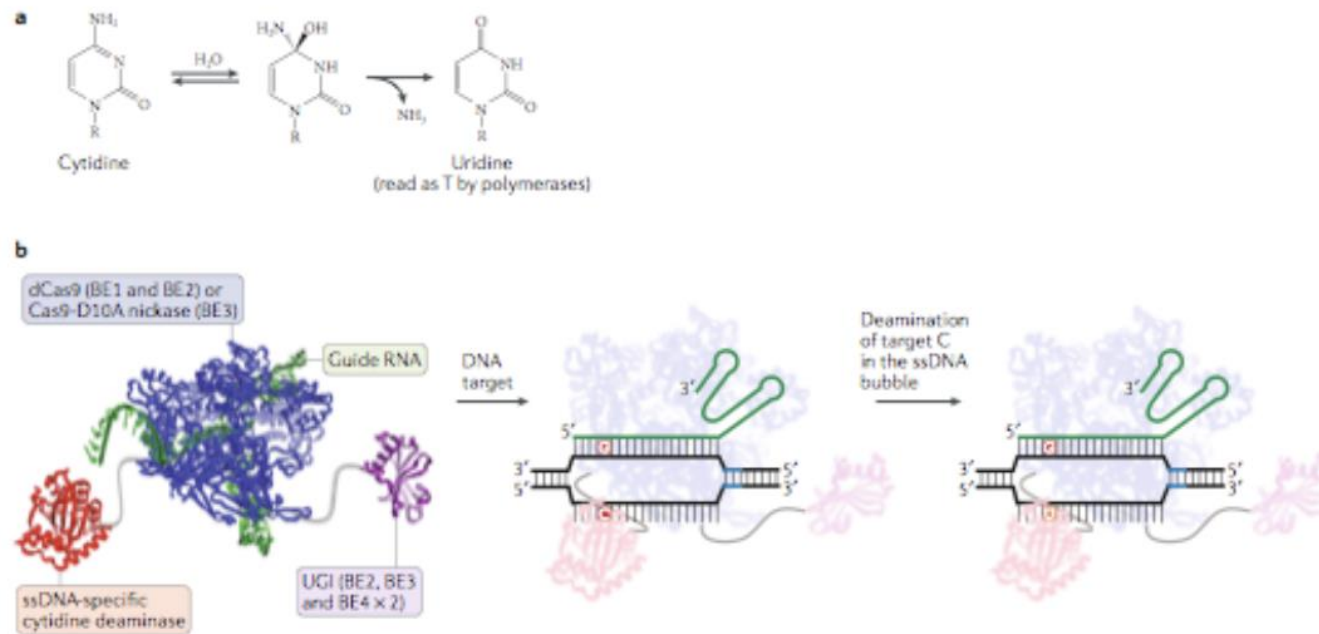


Erwood *et al*, *Nature Biotechnology*, 2022

Coelho *et al*, *Nature Communications*, 2022

Base editing

Base Editing (Cytosine; CBE)

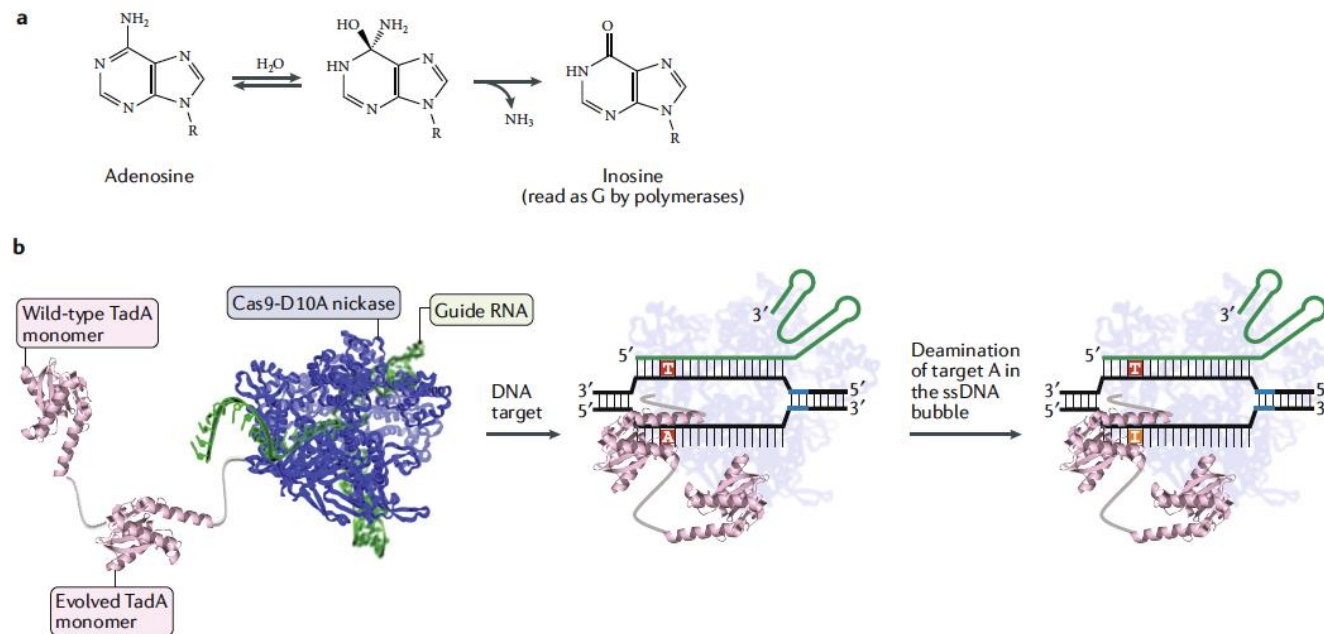


**Available from Addgene
#158581**

Rees and Liu, Nature Reviews Genetic (2018)

Base editing

Base Editing (Adenine; ABE)

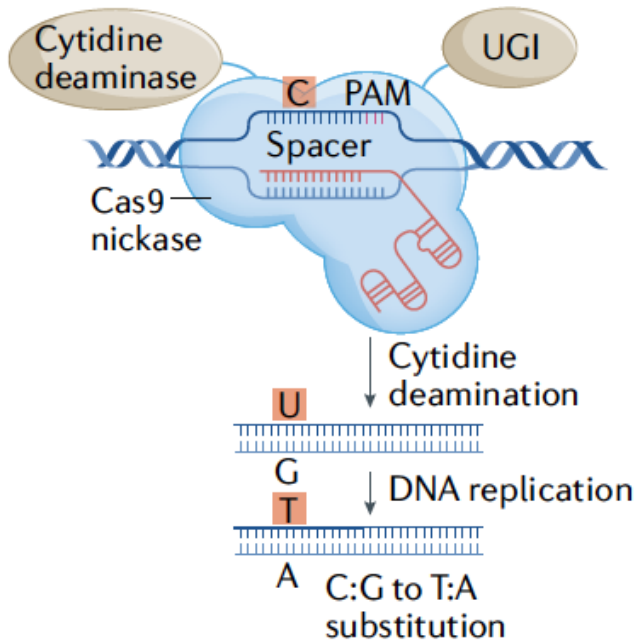


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Rees and Liu, Nature Reviews Genetic (2018)

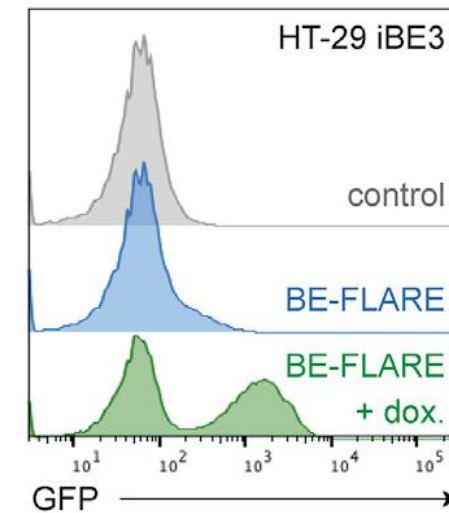
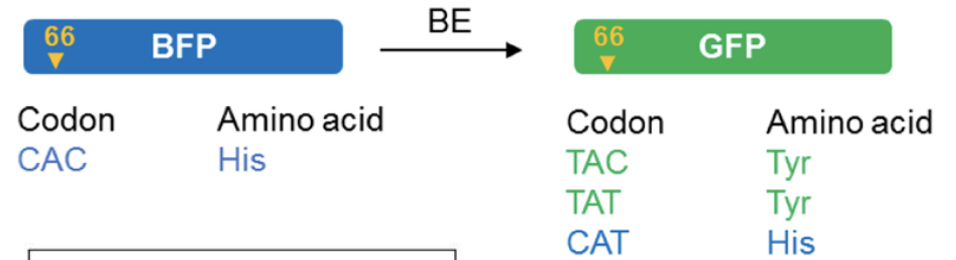
Base editing

CRISPR base editing



Bock, Datlinger, Chardon, Coelho *et al*,
Nature Reviews Methods Primer, 2022

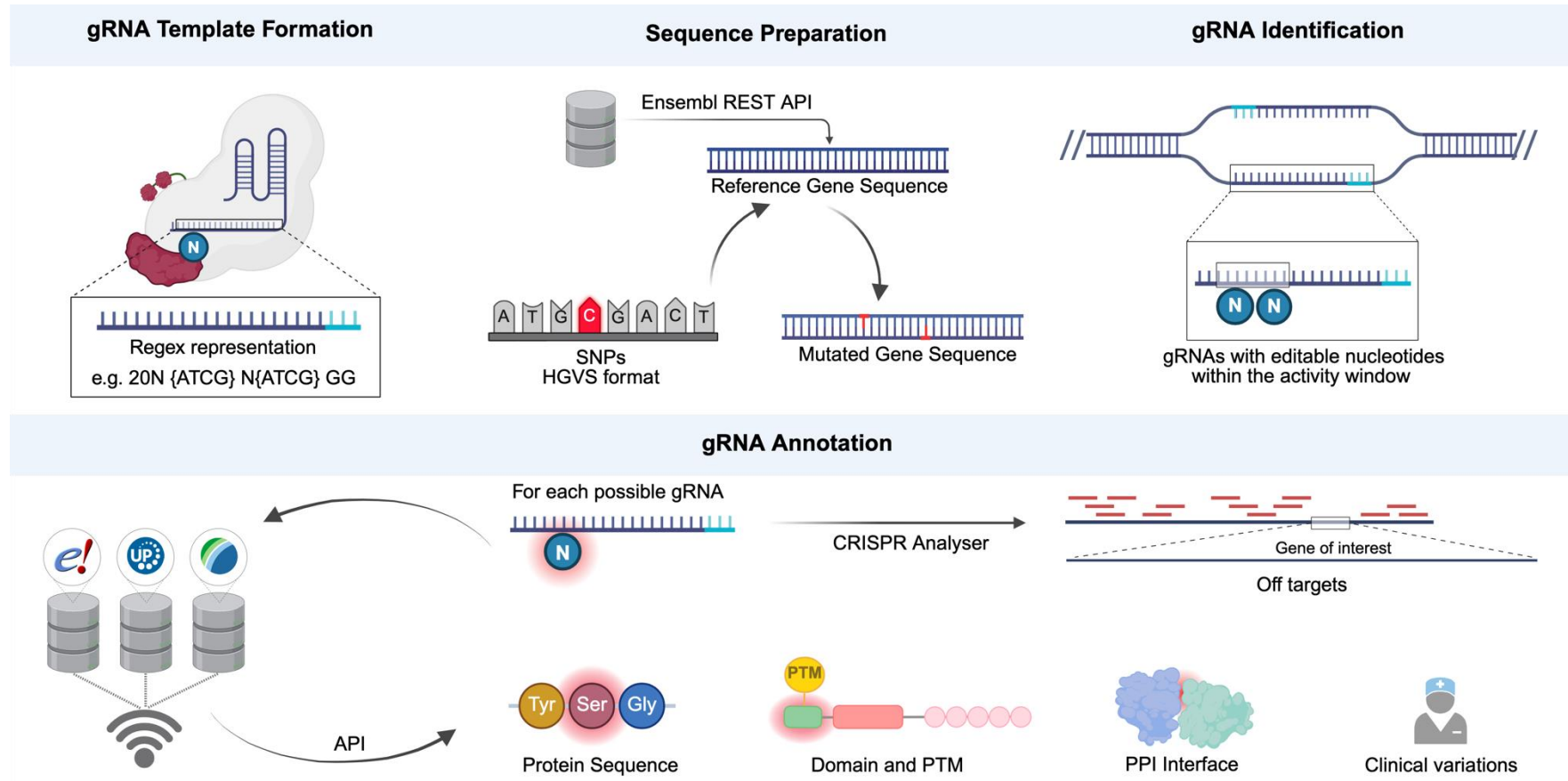
BE-FLARE base editing activity reporter



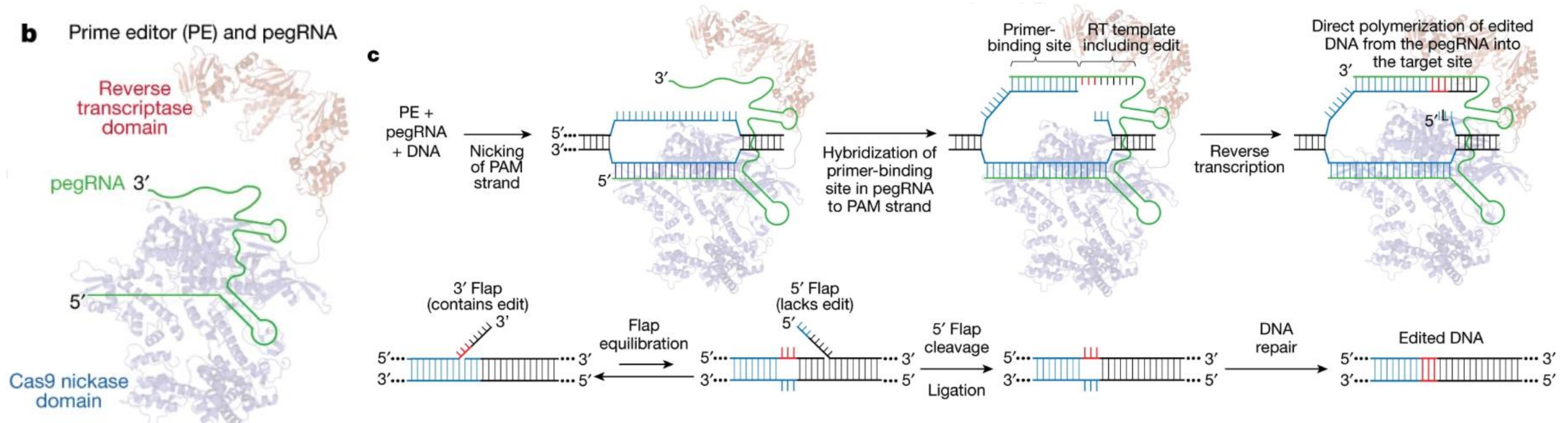
Coelho* *et al*, BMC Biology, 2018

A tool for the design and analysis of base editing screens

*B*Estimate



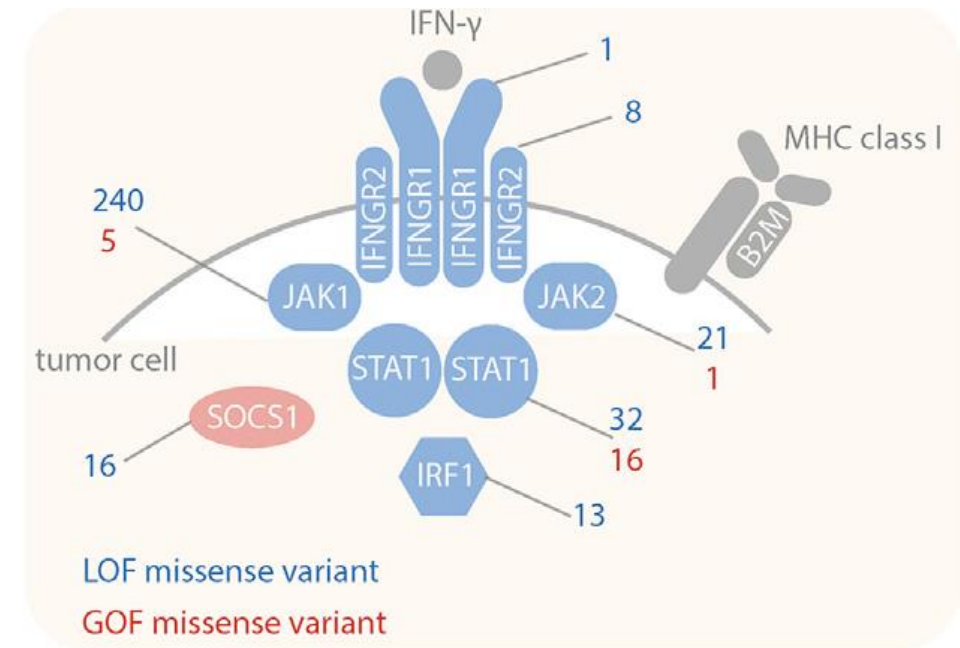
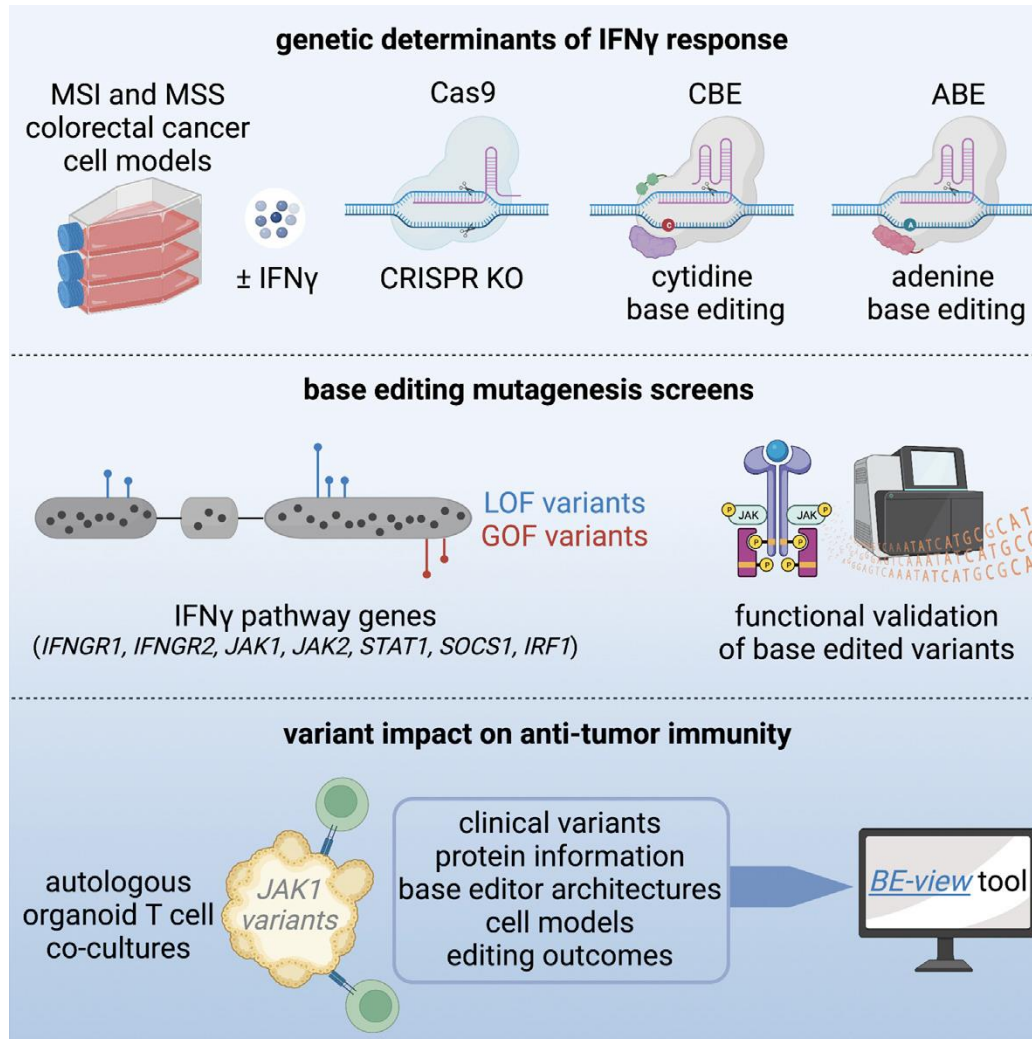
Prime editing



Available from Addgene
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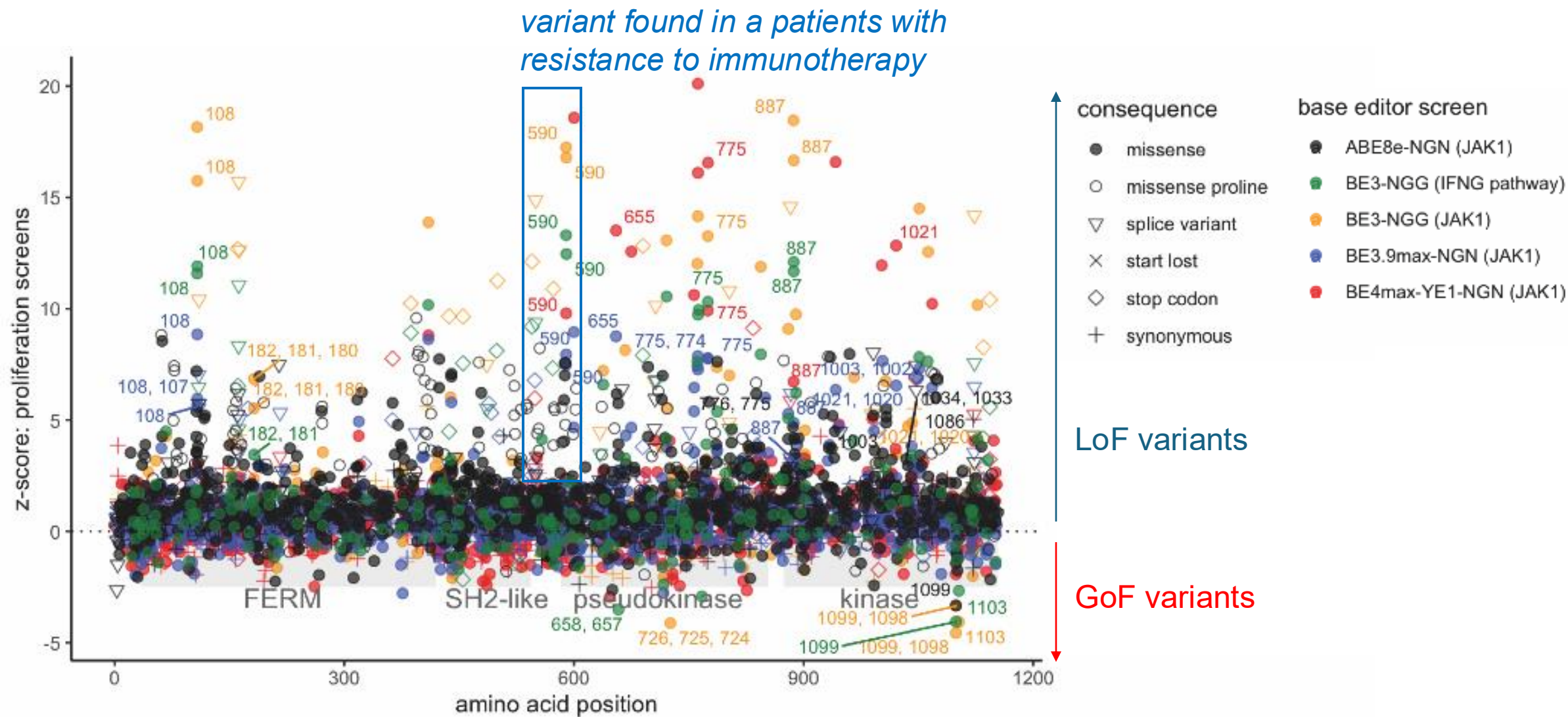
Anzalone, *Nature*, 2019

Pathway-level interrogation of variants affecting IFN- γ signaling

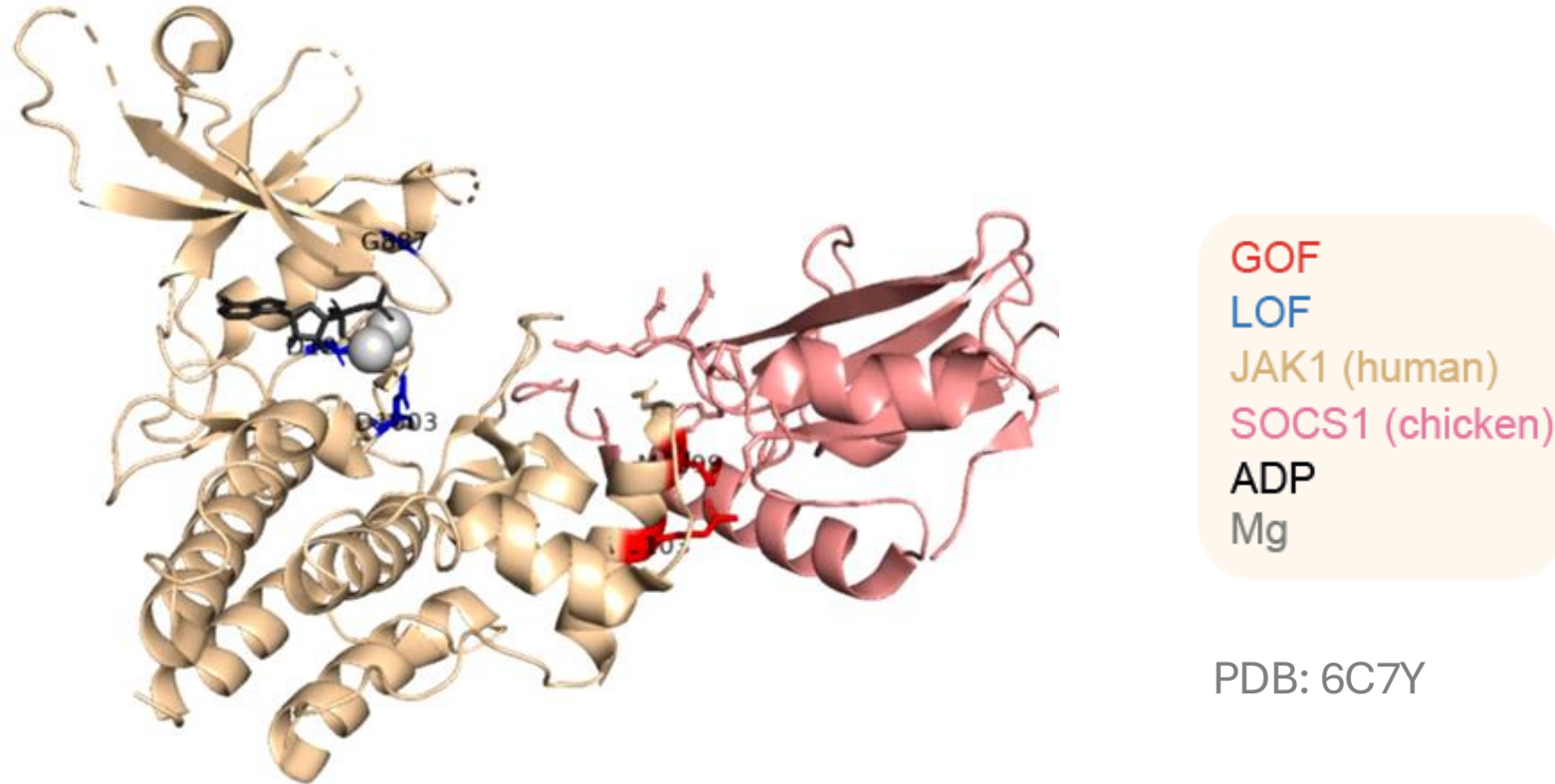


Coelho et al, *Cancer Cell*, 2023

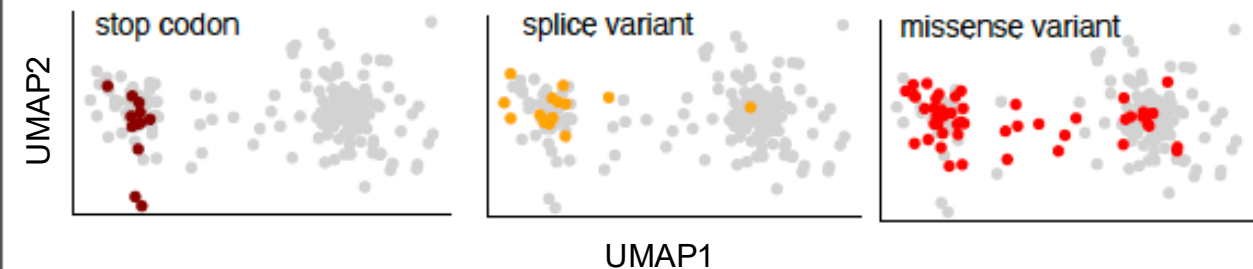
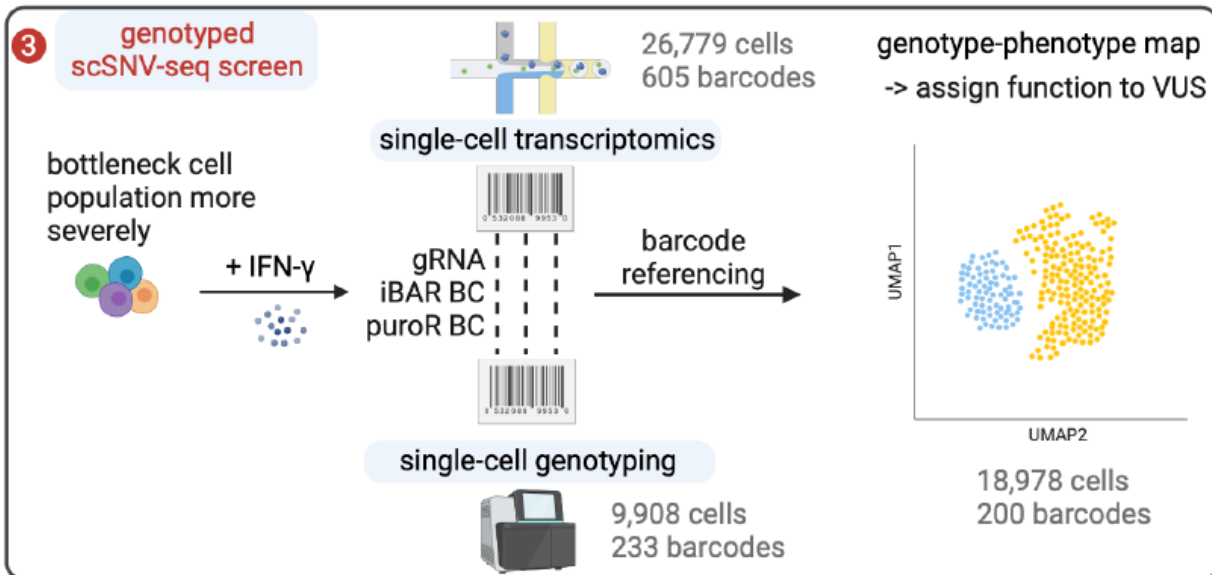
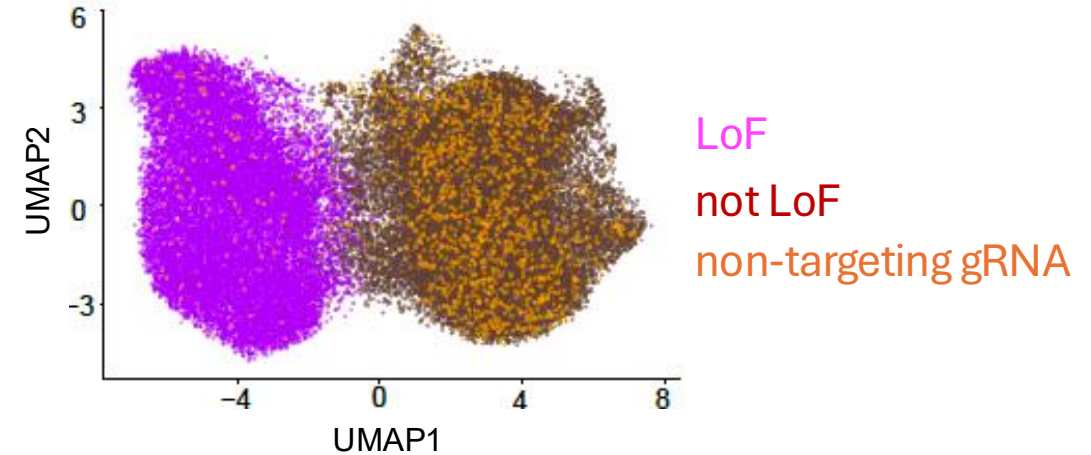
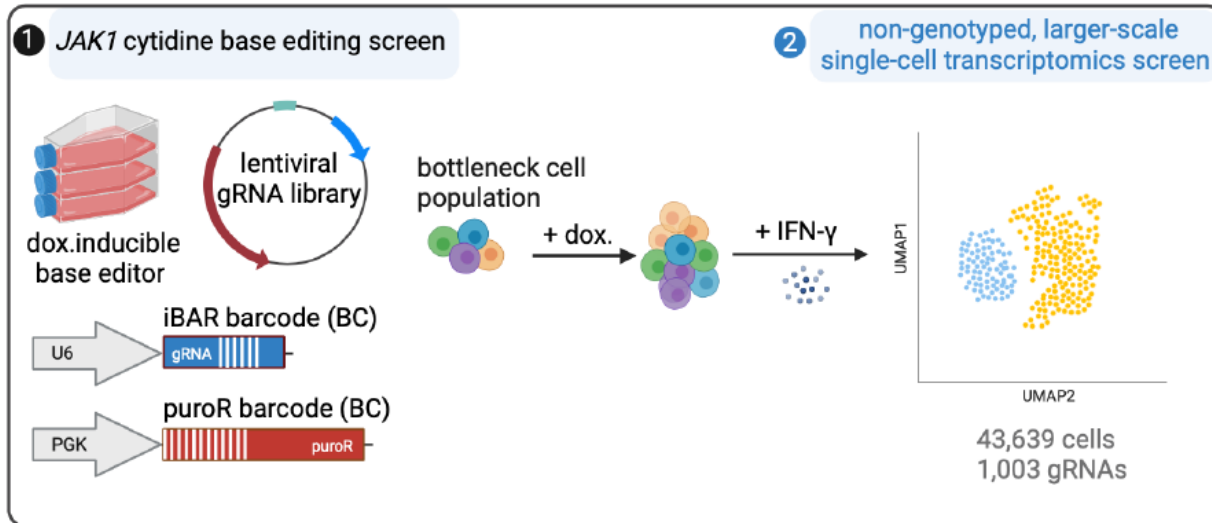
Deep mutagenesis of *JAK1* using base editing



Mechanism of action of JAK1 LoF and GoF variants

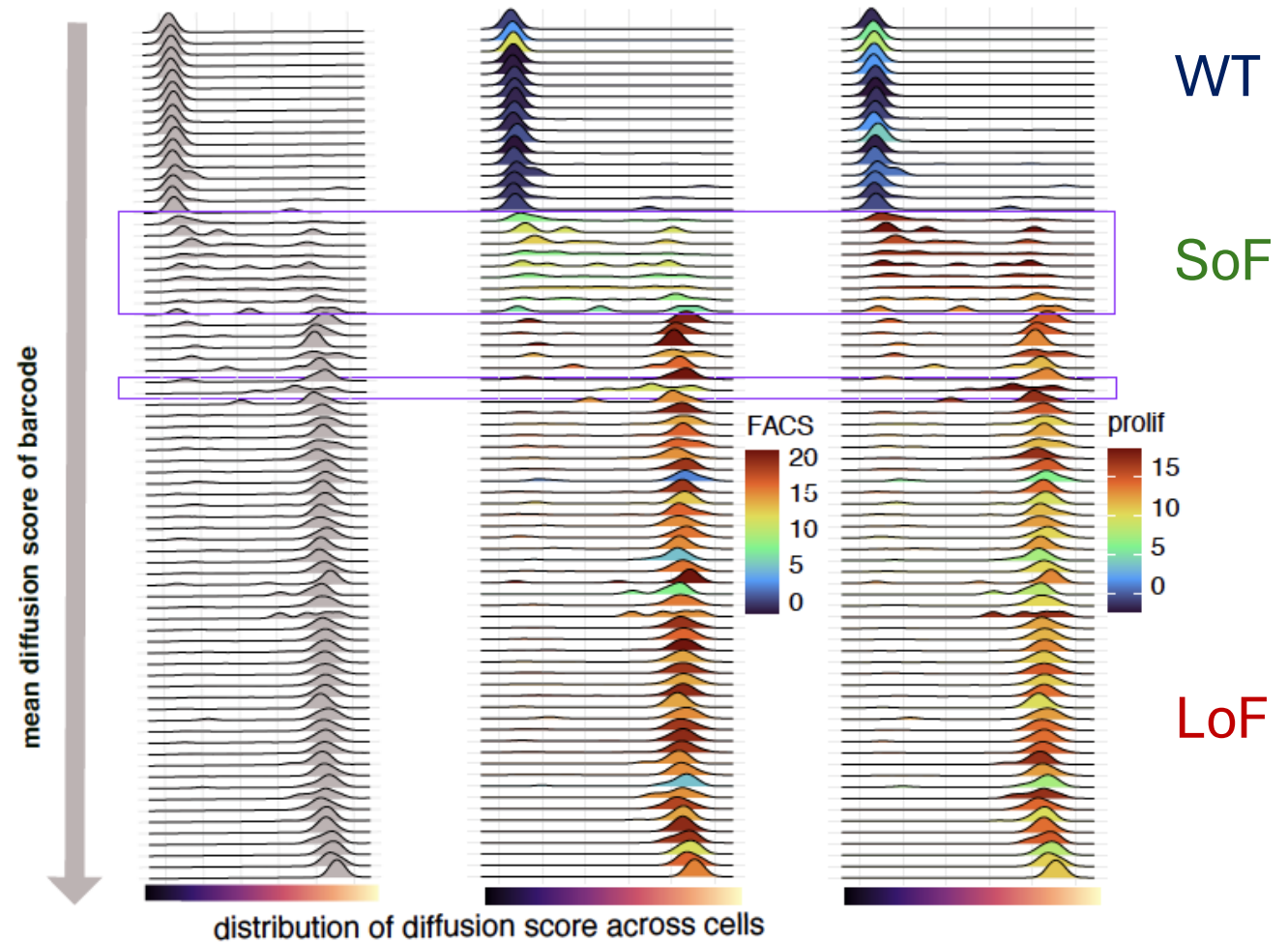
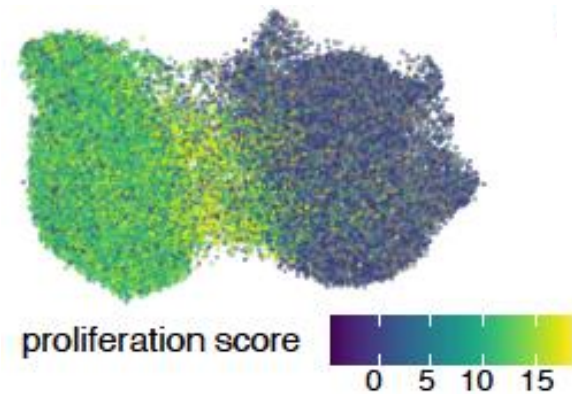
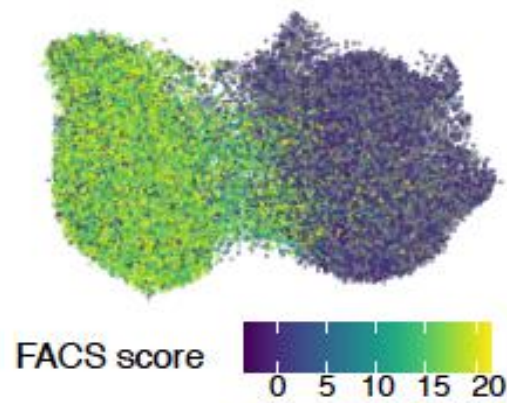


Transcriptomic readouts of variant effects in single cells

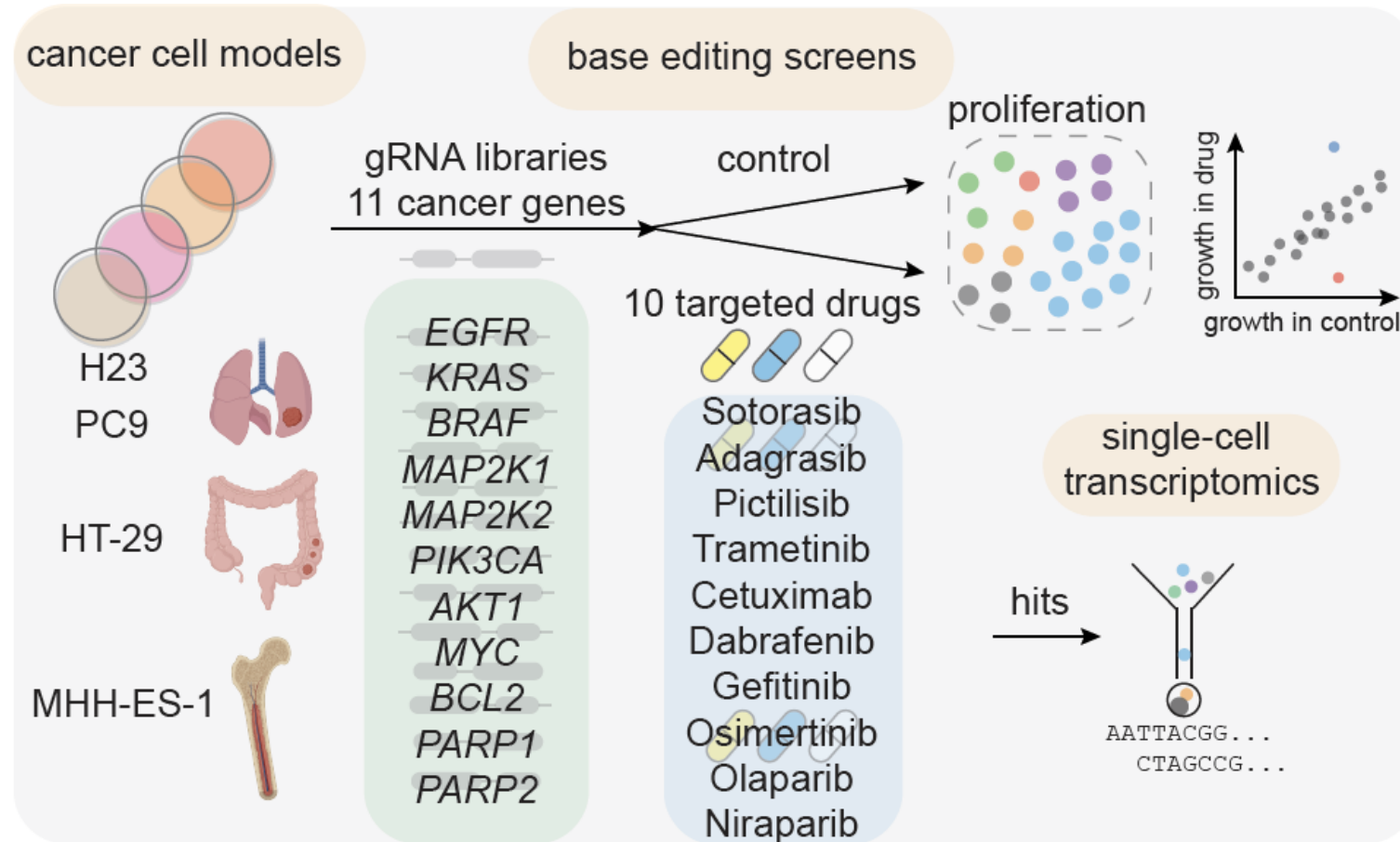


Cooper[‡], Coelho[‡], Strauss[‡] *et al*, *Genome Biology*, 2024

scSNV-seq reveals cellular transcriptional heterogeneity

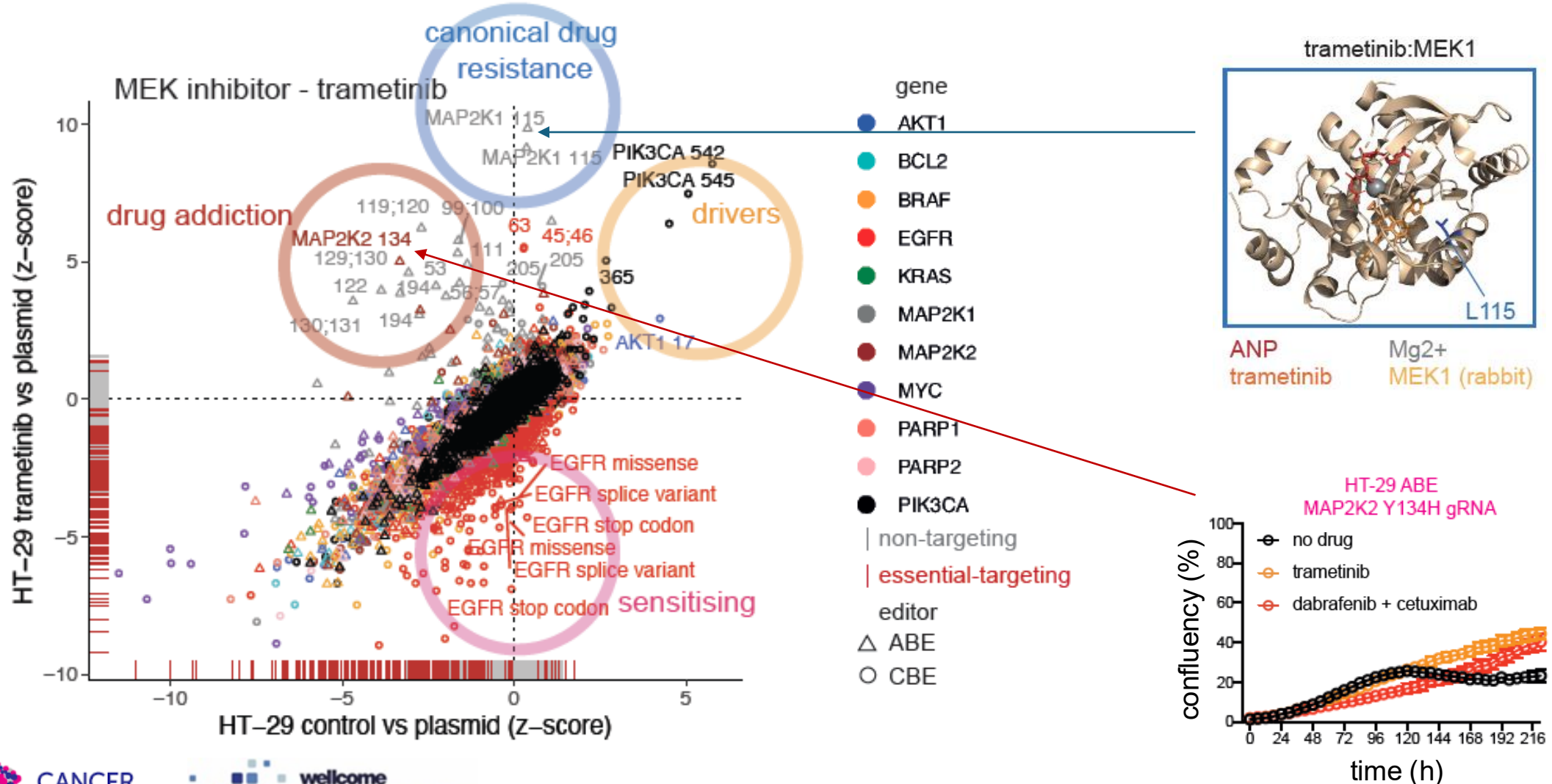


Base editing defines the genetic landscape of cancer drug resistance mechanisms

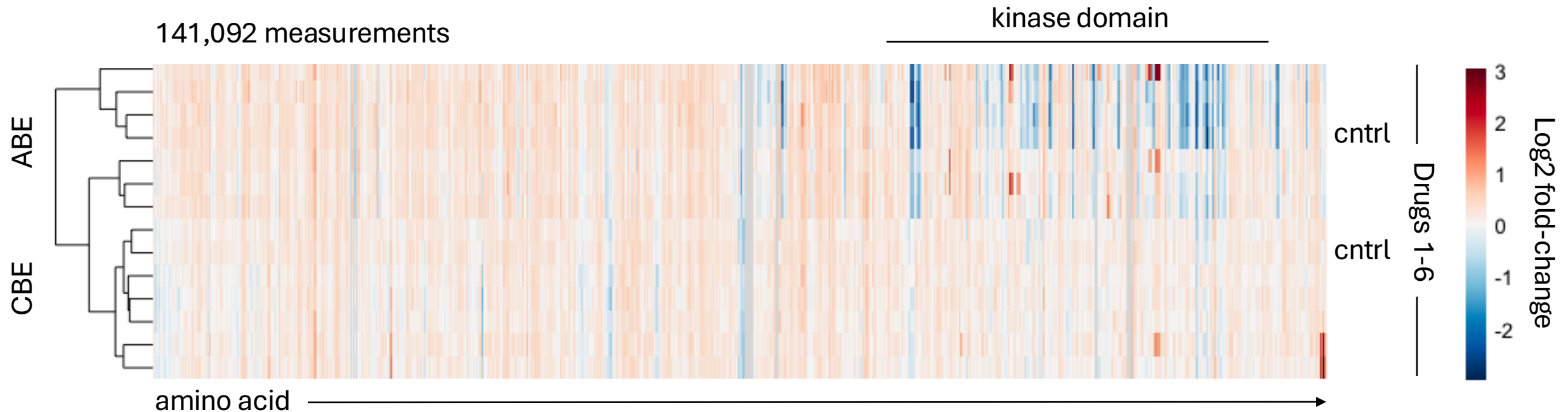


Coelho et al, *Nature Genetics*, 2024

Variants modulating drug sensitivity cluster into four functional classes

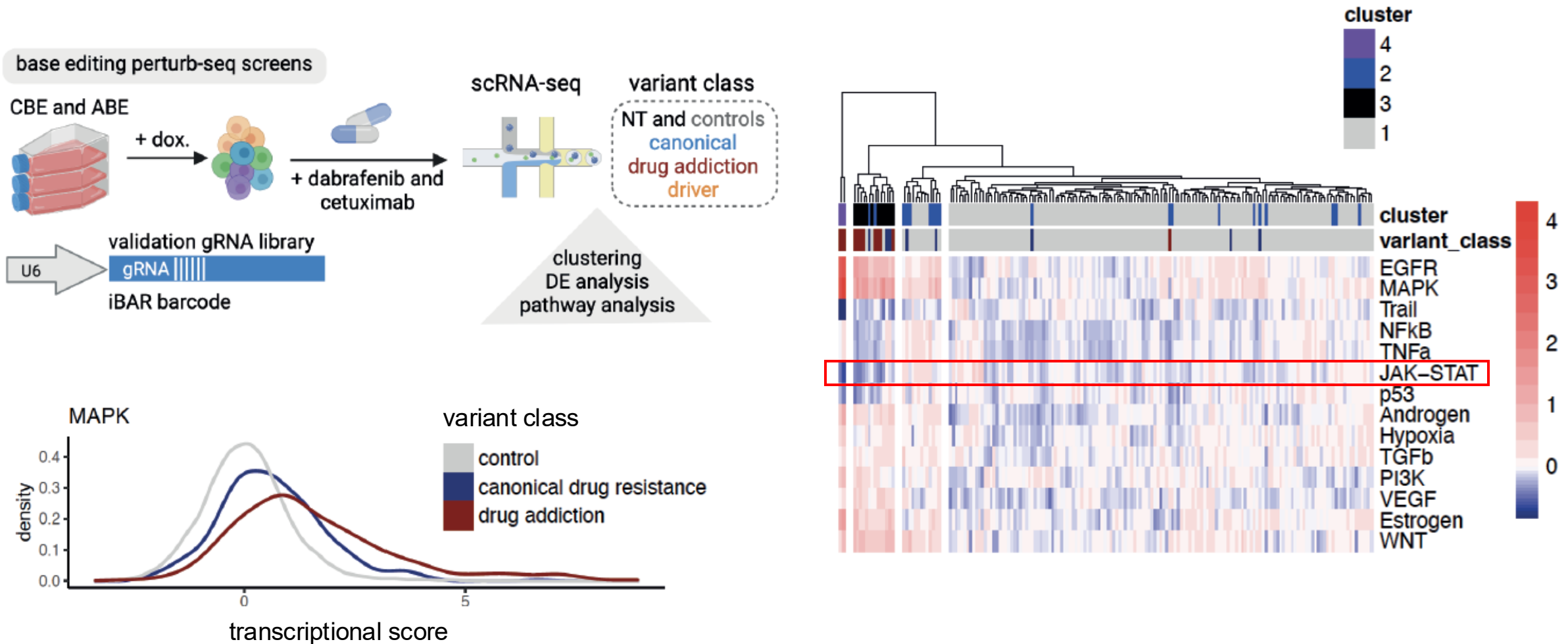


Map of differential sensitivity to NTRK1 inhibitors



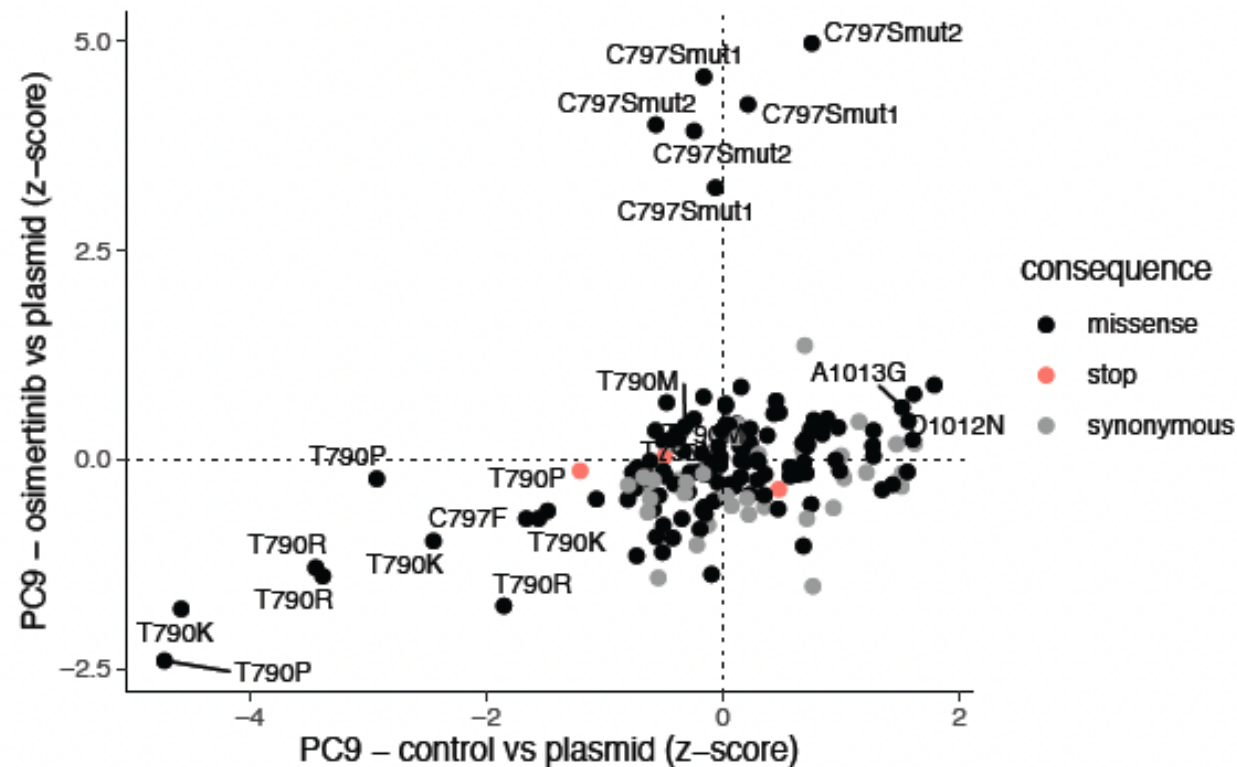
unpublished

Single-cell RNA seq functionally defines drug resistant cell states

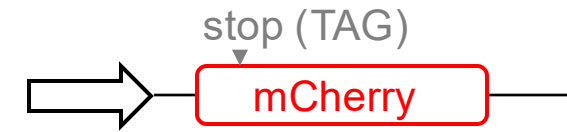


Prime editing screens of cancer variant function

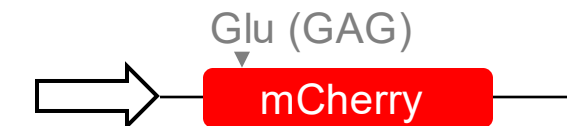
EGFRi osimertinib resistance



reporter of prime editing

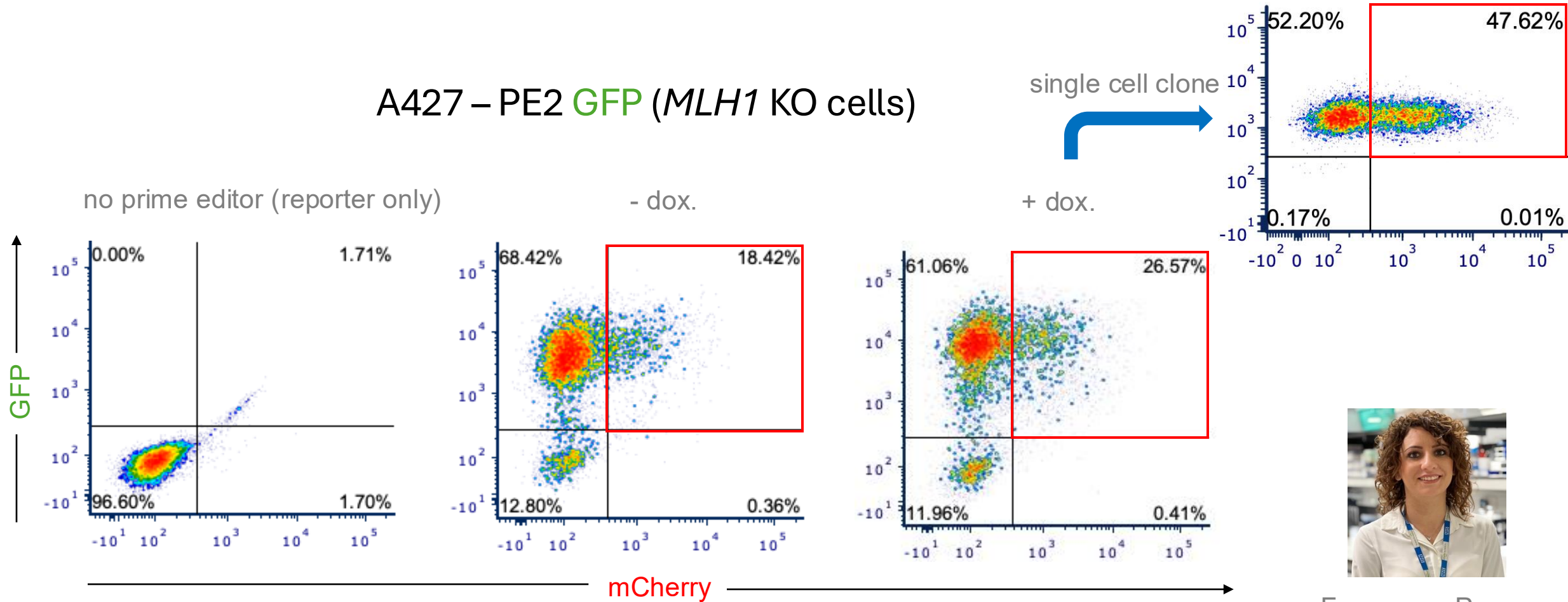


productive prime editing (mU6 -> epegRNA)



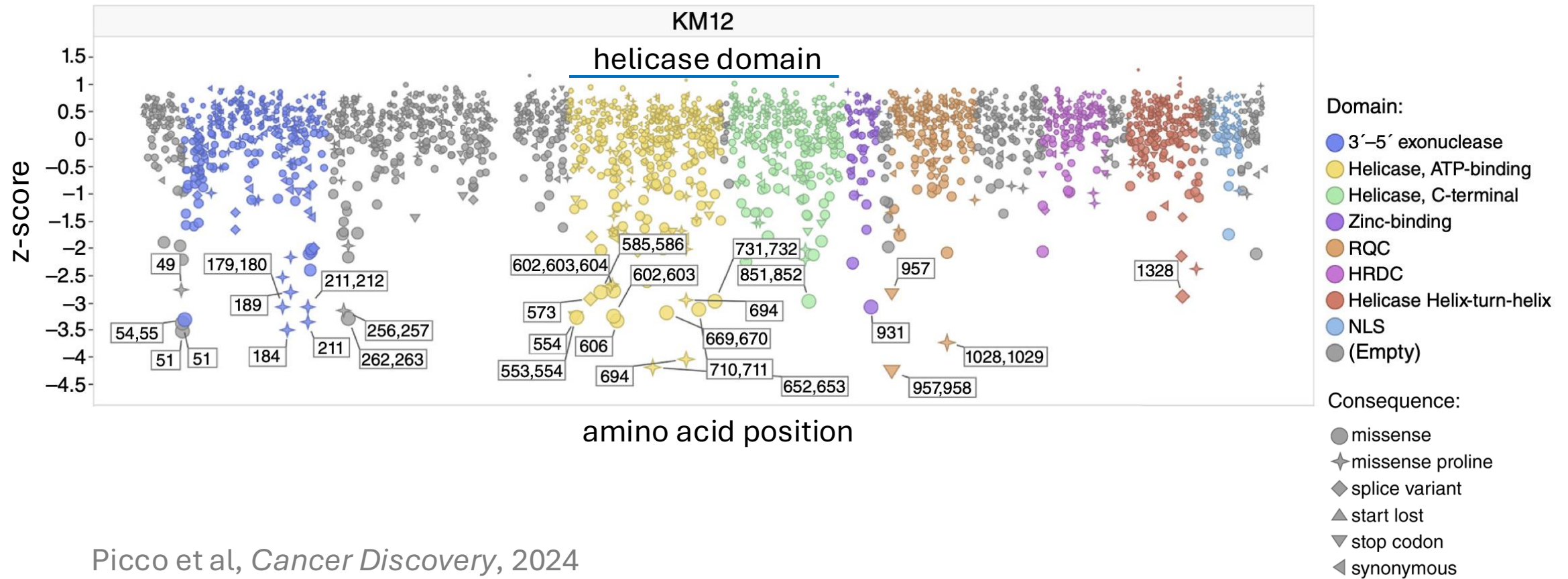
A fluorescent reporter of prime editing activity enables enrichment of edited cells

A427 – PE2 **GFP** (*MLH1* KO cells)



Francesca Perrone

Mutagenesis screens to guide drug design



Picco et al, *Cancer Discovery*, 2024

Somatic variant interpretation

Key challenges:

- Often directly related to clinical actionability, but variant effect data in the presence of a therapy is very limited and slow to accumulate (e.g. COSMIC)
- Limited curated ground truth data on drug resistance variants, and typically only for the most common variants and the oldest drugs
- Almost no ground truth control data on “benign” variants in the presence of therapy

Summary

Important use cases for gene editing MAVEs in cancer treatment:

1. Rapidly identify drug resistant variants (e.g. targeted drugs)
2. Nominate effective second-line therapies and scheduling (e.g. NTRK1)
3. Guide *de novo* drug design (e.g. WRN)

Acknowledgements

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BASE Rx

